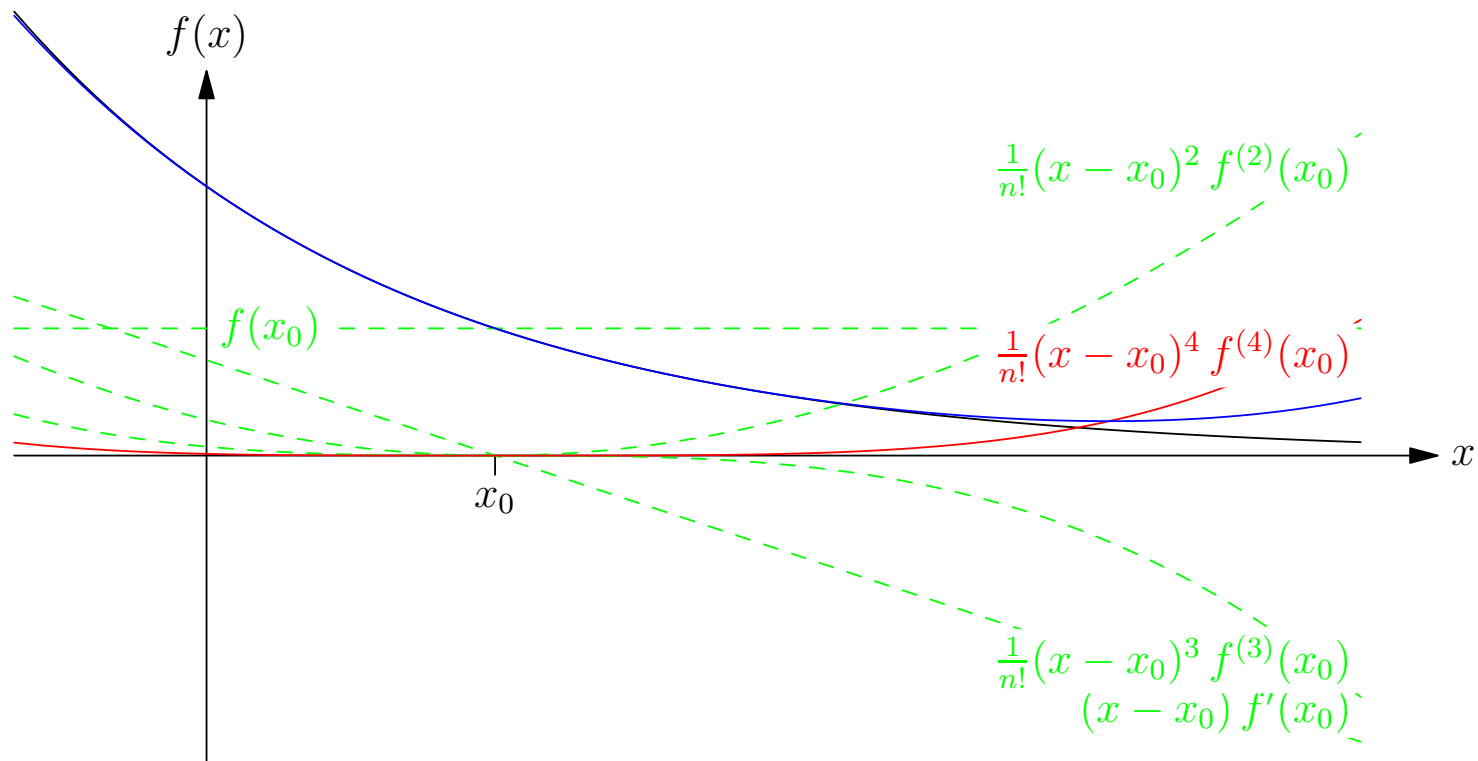


Mathematics of Machine Learning

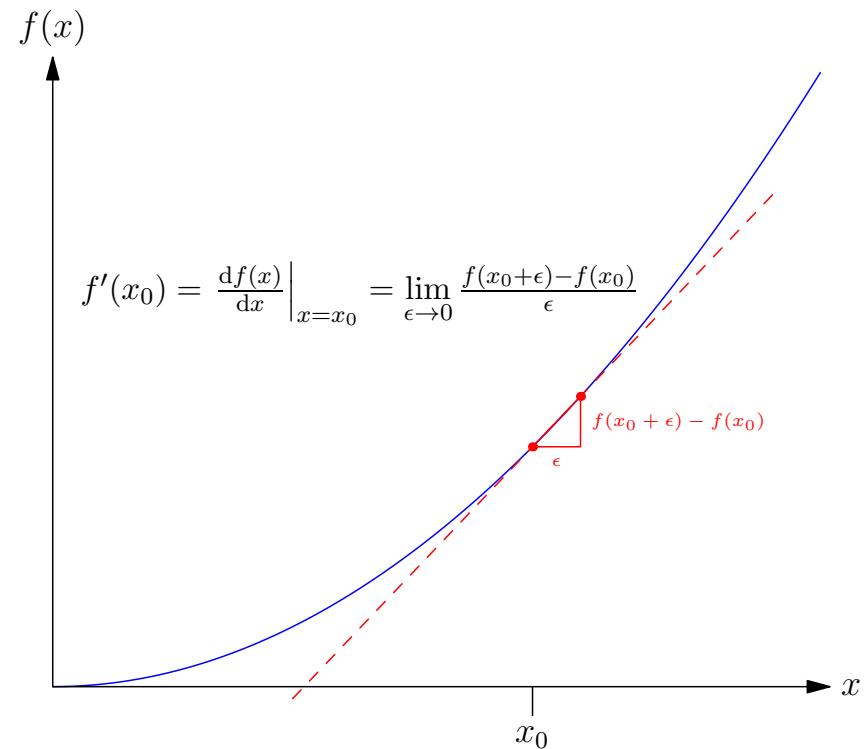
Differential Calculus



Differentiation, product and chain rules, vectors and matrices

Outline

1. **1-D Taylor Expansion?**
2. n -D Taylor Expansion?



Approximating Functions

- In much of machine learning we are trying to approximate functions■
- There are many different ways to do this■
- Depending on your application some are much better than others■
- All function approximation makes some assumptions about the regularity of the function they are approximating■
- Taylor expansions are polynomial approximations of a smooth function around a point■

Polynomial Approximation

- Suppose we wanted to approximate a function $f(x)$ around a point x_0 using a polynomial

$$f(x) \approx a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots + a_n(x - x_0)^n$$

- Taking derivatives

$$f'(x) \approx a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + \dots + na_n(x - x_0)^{n-1}$$

$$f''(x) \approx 2a_2 + 6a_3(x - x_0) + \dots + n(n-1)a_n(x - x_0)^{n-2}$$

- One way to decide on the coefficients is to set $x = x_0$ and assume equality

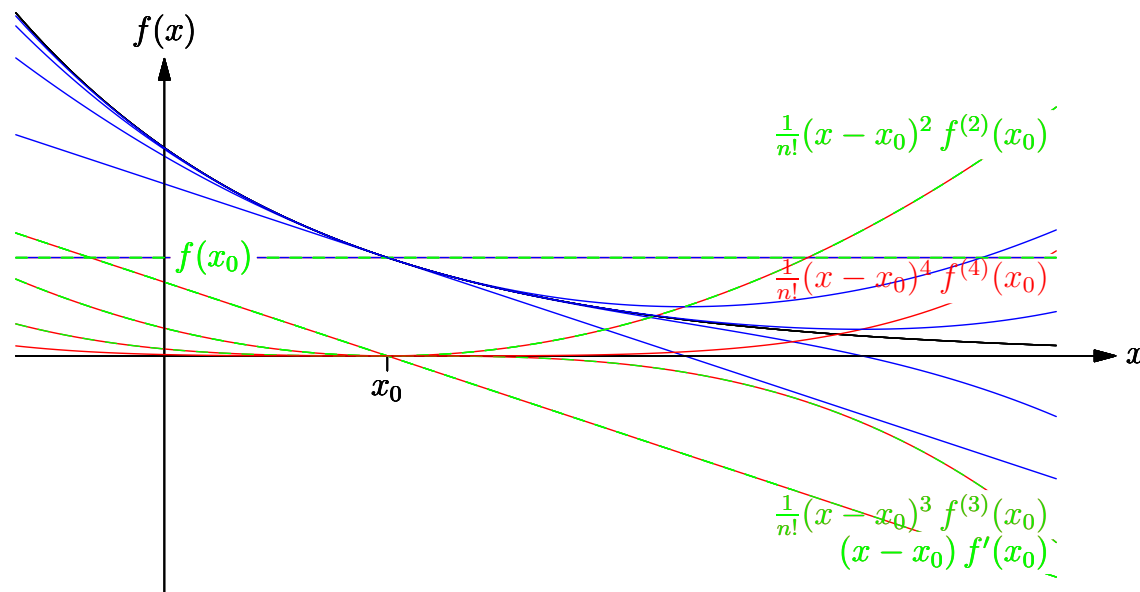
$$a_0 = f(x_0), \quad a_1 = f'(x_0), \quad a_2 = \frac{1}{2}f''(x_0), \quad a_i = \frac{1}{i!}f^{(i)}(x_0)$$

Taylor Expansion

- This gives the Taylor expansion

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + \dots + \frac{1}{n!}f^{(n)}(x_0)(x - x_0)^n$$

- The left and right-hand side both have the same n derivatives at the point x_0



Convergence

- As we add more and more terms our local approximation becomes more accurate, provided the higher derivatives exist
- Polynomials have the nice property that they only diverge at ∞
- However, polynomials cannot capture functions that diverge for finite numbers, e.g.

$$\frac{1}{x}$$

$$\frac{1}{(x - 5)^2}$$

$$\tan(x)$$

- This leads to a loss of global convergence

Taylor Expansion of $\tan(x)$

